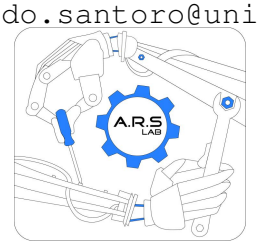
The ARX Model

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Advanced Robotics and Autonomous Systems

Notes

Standard Model from Linear Differential Equation

Standard Model (State Space) from Linear Differential Equation

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

Let's reorder

$$\begin{aligned} \dot{x_1} &= x_2 \\ \dot{x_2} &= -0.5x_1 - 3x_2 + 4u \end{aligned}$$

And put system sign:

$$\begin{cases} \dot{x_1} &= x_2 \\ \dot{x_2} &= -0.5x_1 - 3x_2 + 4u \end{cases}$$

Notes

Standard Model from Linear Differential Equation

Let's put in standard matrix form

$$\begin{cases} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -0.5x_1 - 3x_2 + 4u \end{cases}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 \\ -0.5 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

So we have:

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}$$

Notes

Let's Discretize

The system:

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx \end{cases}$$

becomes:

$$\begin{cases} x(k+1) &= \widetilde{A}\, x(k) + \widetilde{B}\, u(k) \\ y(k) &= \widetilde{C}\, x(k) \end{cases}$$

here:

$$\begin{aligned} \widetilde{A} &= A\Delta T + I \\ \widetilde{B} &= B\Delta T \\ \widetilde{C} &= C \end{aligned}$$

Notes

Standard Model from Linear Differential Equation

Thus:

$$\begin{cases} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -0.5x_1 - 3x_2 + 4u \end{cases}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 \\ -0.5 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

By applying discretization, we have:

$$\tilde{A} = \begin{bmatrix} 1 & \Delta T \\ -0.5\Delta T & -3\Delta T + 1 \end{bmatrix} \quad \tilde{B} = \begin{bmatrix} 0 \\ 4\Delta T \end{bmatrix} \quad \tilde{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Notes

Discretization from Linear Differential Equation

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

What does it happen if we apply discretization **directly to the differential equation?**

we must apply the following substitutions:

$$\frac{y(k) - y(k-1)}{\Delta T} \simeq \dot{y}$$
$$\frac{\dot{y}(k) - \dot{y}(k-1)}{\Delta T} \simeq \ddot{y}$$

Thus:

$$\frac{\frac{y(k)-y(k-1)}{\Delta T} - \frac{y(k-1)-y(k-2)}{\Delta T}}{\Delta T} \simeq \ddot{y}$$

Notes

Discretization from Linear Differential Equation

Discretization from Linear Differential Equation

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

What does it happen if we apply discretization **directly to the differential equation?**

we must apply the following substitutions:

$$\frac{y(k) - y(k-1)}{\Delta T} \simeq \dot{y}$$

$$\frac{y(k) - y(k-1)}{\Delta T^2} - \frac{y(k-1) - y(k-2)}{\Delta T^2} \simeq \ddot{y}$$

Thus:

$$\frac{y(k) - y(k-1)}{\Delta T^2} - \frac{y(k-1) - y(k-2)}{\Delta T^2} + 3\frac{y(k) - y(k-1)}{\Delta T} + 0.5y(k) = 4u(k)$$

Notes

Discretization from Linear Differential Equation

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

$$\begin{aligned} \frac{y(k) - y(k-1)}{\Delta T^2} - \frac{y(k-1) - y(k-2)}{\Delta T^2} & + \\ +3\frac{y(k) - y(k-1)}{\Delta T} + 0.5y(k) & = 4u(k) \end{aligned}$$

Let's go on:

$$\begin{aligned} y(k) - y(k-1) - y(k-1) + y(k-2) & + \\ +3(y(k) - y(k-1))\Delta T + 0.5y(k)\Delta T^2 & = 4u(k)\Delta T^2 \end{aligned}$$

$$\begin{aligned} y(k) - 2y(k-1) + y(k-2) + 3y(k)\Delta T & \\ -3y(k-1)\Delta T + 0.5y(k)\Delta T^2 & = 4u(k)\Delta T^2 \end{aligned}$$

Notes

Discretization from Linear Differential Equation

Discretization from Linear Differential Equation

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

$$\begin{aligned} y(k) - 2y(k-1) + y(k-2) + 3y(k)\Delta T & \\ -3y(k-1)\Delta T + 0.5y(k)\Delta T^2 &= 4u(k)\Delta T^2 \end{aligned}$$

Let's solve:

$$\begin{aligned} (1 + 3\Delta T + 0.5\Delta T^2)y(k) & + \\ (-2 - 3\Delta T)y(k-1) + y(k-2) &= 4u(k)\Delta T^2 \end{aligned}$$

That becomes:

$$\begin{aligned} y(k) - \frac{2 + 3\Delta T}{1 + 3\Delta T + 0.5\Delta T^2}y(k-1) & + \\ \frac{1}{1 + 3\Delta T + 0.5\Delta T^2}y(k-2) &= \frac{4\Delta T^2}{1 + 3\Delta T + 0.5\Delta T^2}u(k) \end{aligned}$$

Notes

Discretization from Linear Differential Equation

$$\ddot{y} + 3\dot{y} + 0.5y = 4u$$

Final update equation:

$$y(k) = \frac{2 + 3\Delta T}{1 + 3\Delta T + 0.5\Delta T^2} y(k-1) \quad - \quad \frac{1}{1 + 3\Delta T + 0.5\Delta T^2} y(k-2) \quad + \quad \frac{4\Delta T^2}{1 + 3\Delta T + 0.5\Delta T^2} u(k)$$

The current value of the output $y(k)$ depends on:

- The **previous** value of the output $y(k-1)$
- The **value at 2 steps before** of the output $y(k-2)$
- The **current value** of the input $u(k)$

Notes

The ARX Model of a Discrete System

The ARX Model

A discrete system can be modeled as:

$$y(k) + a_1 y(k-1) + a_2 y(k-2) + \cdots + a_n y(k-n) = b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

It is called Auto-Regressive with eXogenous Input.

The output $y(k)$ depends on:

- past values of the output (auto-regressive part)
- current and past values of the input (exogenous input)

Parameters:

- n is the order of the AR part
- m is the order of the X part
- when the model comes from a differential equation, n is the same order in terms of y derivatives

Notes

The ARX Model of a Discrete System

Block Schematic of an ARX Model

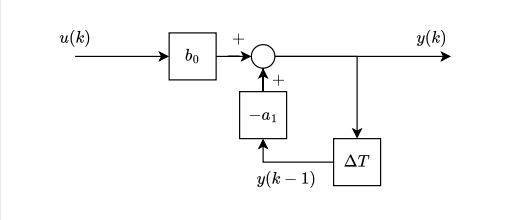
An ARX model:

$$y(k) + a_1 y(k-1) + a_2 y(k-2) + \cdots + a_n y(k-n) = b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

can be drawn as feedback of delay blocks because each term with $k - i$ represents a **delay** of $i\Delta T$.

Example (first order)

$$y(k) = -a_1 y(k-1) + b_0 u(k)$$



Notes

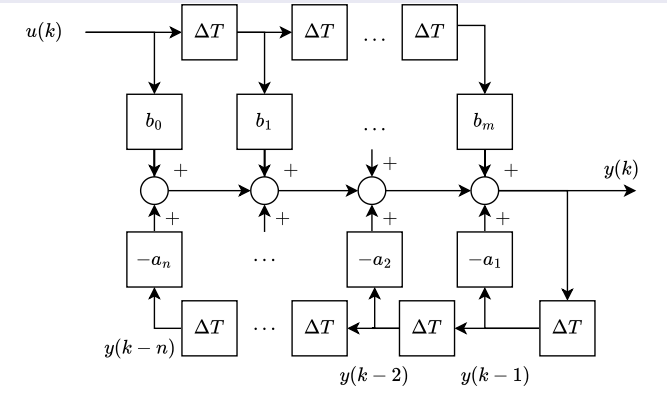
The ARX Model of a Discrete System

Block Schematic of an ARX Model

An ARX model:

$$y(k) + a_1 y(k-1) + a_2 y(k-2) + \cdots + a_n y(k-n) = b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

can be drawn as feedback of delay blocks because each term with $k - i$ represents **delay** of $i\Delta T$.



Notes

From ARX to State Space

Notes

Standard Model from ARX

Standard Model (State Space) from ARX

Given:

$$y(k) + 3y(k-1) + 0.5y(k-2) = 4u(k)$$

how can we derive the state-space model?

$$x(k+1) = Ax(k) + Bu(k)$$

The approach is similar to that of continuous systems

Notes

Standard Model from ARX

Standard Model from ARX

$$y(k) + 3y(k-1) + 0.5y(k-2) = 4u(k)$$

Rules (by example)

- Without loss of generality, we can shift the output in time:

$$y(k+2) + 3y(k+1) + 0.5y(k) = 4u(k)$$

- Let's assume:

$$\begin{aligned} x_1(k) &= y(k+1) \\ x_2(k) &= y(k) \end{aligned}$$

Notes

Standard Model from ARX

Standard Model from ARX

$$y(k+2) + 3y(k+1) + 0.5y(k) = 4u(k)$$

Rules (by example)

- Let's assume:

$$\begin{aligned} x_1(k) &= y(k+1) \\ x_2(k) &= y(k) \end{aligned}$$

- We have:

$$\begin{aligned} x_1(k+1) &= y(k+2) \\ x_2(k+1) &= y(k+1) = x_1(k) \end{aligned}$$

Notes

Standard Model from ARX

Standard Model from ARX

$$y(k+2) + 3y(k+1) + 0.5y(k) = 4u(k)$$

Substitution

$$\begin{aligned} x_1(k+1) &= y(k+2) \\ x_2(k+1) &= y(k+1) = x_1(k) \\ x_2(k) &= y(k) \end{aligned}$$

Substitution

$$\begin{aligned} x_1(k+1) + 3x_1(k) + 0.5x_2(k) &= 4u(k) \\ x_2(k+1) &= x_1(k) \end{aligned}$$

Notes

Standard Model from ARX

Standard Model from ARX

$$y(k+2) + 3y(k+1) + 0.5y(k) = 4u(k)$$

Let's reorder

$$\begin{cases} x_1(k+1) &= -3x_1(k) - 0.5x_2(k) + 4u(k) \\ x_2(k+1) &= x_1(k) \end{cases}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad A = \begin{bmatrix} -3 & -0.5 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

So we have:

$$\begin{aligned} x(k+1) &= Ax + Bu \\ y &= Cx \end{aligned}$$

Notes

Eigenvalues and Stability

Given a linear discrete system defined by

$$x(k+1) = Ax + Bu$$

given that the eigenvalues of A are:

$$\lambda_i \in \mathcal{C} \qquad (\text{remind that } \mathcal{R} \subset \mathcal{C})$$

i.e. λ_i may be either **real** or **complex and conjugate**, we have

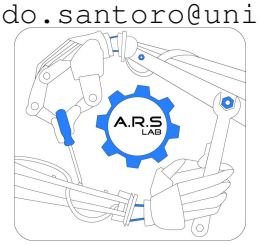
Condition	System Stability
All $\lvert\lambda_i\rvert < 1$	**Asymptotical**
At least one $\lvert\lambda_{i*}\rvert = 1$, but, if λ_{i*} is compl-and-conj, it is simple (mutiplicity = 1)	**Simple**
All other cases	**Instability**

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